Stackly Courier: technical notes

Stack

Vanilla HTML, CSS and JS. There are no frameworks, no CDNs and no external requests. The pages are generated by Python (`build.py`), so the header and footer are already in the markup before any JS runs. There is one shared `styles.css` and one shared `script.js`. Fonts are self-hosted WOFF2: Cormorant Garamond for headings and Manrope for body text. `fonts.py` inlines them into `styles.css` as data URIs, so they also load when a page is opened straight from disk (file://), where browsers block separate font files. The latin-ext files are cut down to the ₹ sign only.

Layout

The deploy folder is flat and ready for GitHub Pages under a subfolder path, because every link is relative.

- `*.html` (13 pages), `styles.css`, `script.js`
- All images (`*.webp`, `favicon.svg`) sit in the same folder as the pages.
- No separate font folder: the fonts are inlined in `styles.css`.

Footer and navigation

- The header menu, mobile menu and footer Navigate column all come from one `NAV` list: Home, Services, Track, Pricing, About, Contact.
- Footer social links are icon buttons (LinkedIn, Instagram, X) under the logo, each with an aria-label. Like the footer phone and email, they go to 404.html.

Dropdowns, footer legal links and dashboard user

- Every `<select>` is enhanced into a custom dropdown (`initSelects`). The option list is anchored to the field and is never wider than it, so it cannot run off small screens. The real `<select>` stays in the page for values and validation. Arrow keys, Enter, Space, Escape and Tab all work.
- The footer bottom-right corner has Privacy policy and Terms & conditions links. Both go to 404.html.
- The dashboard header shows the signed-in user's initial, name and email. On screens 760px and narrower, this moves to its own full-width row under the header.
- On Contact, the form column and the photo-and-details column are stretched to the same height.

Tokens and contrast (WCAG ratio): "Midnight & Champagne"

Token	Hex	Pairing checked
onyx	#0B0E13	text #EDE6D8 on onyx 15.57
panel	#171E28	text on panel 13.5, muted #ABA496 on panel 6.77
champagne	#C9A86A	gold on onyx 8.55, onyx on gold 8.55
champagne light	#E6D3A3	on panel 11.34
ivory	#F4EFE6	ink #16191F on ivory 15.37, muted #5A554C on ivory 6.46
gold dark	#7D5F28	on ivory 5.18, on sand 4.70
error	#B42318	with white 6.57, on ivory 5.74

Each section sets surface variables (`--fg`, `--mute`, `--acc`, `--card`, `--line`). Every component inherits a palette that has been checked for contrast against that section's background.

script.js functions

Shared state is declared first: `REAL_PAGES`, `NAV`, `STATE`, `KEYS`, `COMMON_PASSWORDS` and `FX`.

Function	What it does
`sweepLinks`	Sends any href not in REAL_PAGES to 404.html, including tel: and mailto: links (the footer phone and email go to 404 on purpose).
`closeOverlays`	Closes all overlays on pageshow, popstate and pagehide: the menu, drawer, splash, modal, lightbox and notifications.
`initMenu`	Burger menu.
`initSplash`	Logo click shows an opacity-only splash, then goes home.
`initReveals`	IntersectionObserver reveals, staggered. Anything in the first viewport is shown instantly, and a scroll sweep catches anything skipped.
`initFx` / `runFx`	Set-pieces dispatched by `data-fx`: `route`, `count`, `bars` and `timeline`. An unknown value logs `console.error`. A scroll sweep makes sure jump-scrolls never skip a set-piece.
`initTilt`	Hero card tilt, for fine pointers only.
`initCarousel3d`	Hero 3D speed carousel.
`initTestimonials`	Testimonials carousel.
`initLightbox`	About page gallery lightbox.
`initForms`, `V.*`, `HANDLERS.*`	Custom validation and the submit flows.
`initDashboard` views	Each sidebar item shows only its own view (`.view` sections switched by the URL hash, e.g. `#shipments`), with the active link marked.
`initAuth`, `initDashboard`	sessionStorage session, role guard, drawer, bell, log out.
`initServices`, `initTrack`, `initCalc`, `initTabs`	Page-specific features.
`initMotion`	Hero and band parallax, magnetic buttons, card spotlight, header state, and the fade between pages.

With reduced motion on, every set-piece is run to its end state immediately and reveals are shown at once.

Validation rules

- **Name:** letters only ("Only alphabets are allowed."). Space key blocked, spaces stripped on paste, max 16.
- **Phone:** digits only ("Only numbers are allowed."), exactly 10 digits ("Enter exactly 10 digits."), must start with 6–9.
- **Email:** max 64. Rejects whitespace, more than one @, `..`, leading or trailing dots, labels that start or end with a hyphen, single-character labels, all-numeric labels and

non-letter endings.

- **Password:** the common-password list is checked first, then length 8 or more, then uppercase, lowercase, number and symbol, each with its own message. Max 32. Confirm must match.
- **Pincode:** exactly 6 digits, not starting with 0.
- **Tracking ID:** SK followed by 10 digits.
- **Weight:** 0.1–50 kg.
- **Dimensions:** 1–150 cm.
- **Pickup date:** must be after today.
- **Selects and role choice:** "Please choose an option."
- **Textarea:** 10–600 characters.
- **Remember me:** required.
- **Terms:** a popup blocks sign-up if the box is unticked.

Errors show on blur and on submit, and clear as soon as the field is edited. Bubbles use `pointer-events:none`, and the field's margin grows so a bubble never covers the next field.

Rebuild

```
python3 images.py      # crops and compresses photos (needs the source JPGs in
/mnt/user-data/uploads)
python3 logo.py        # makes the light and dark logo cut-outs
python3 build.py       # writes the 13 HTML pages and script.js into ../site
python3 fonts.py       # inlines the fonts into styles.css (run after editing
styles.css)
```

`styles.css` is edited by hand inside the site folder.

Tests

```
pip install playwright && playwright install chromium
SITE_DIR=path/to/site python3 tests/run_tests.py
```

The suite starts its own HTTP server. It fails on any page error or console error. Results are written to `tests/last-results.json`.

Honest limitations

- A 41-character email fits the field at 480px and wider. Narrower than that, the text scrolls inside the field, because shrinking it below 16px would trigger zoom on iOS.
- Accounts are stored in this browser's localStorage, so users can sign in later. The signed-in name is also mirrored to sessionStorage. There is no backend, so an account only exists on the device where it was created.

- Tracking always shows the same sample shipment. The route map and the Contact map are hand-drawn SVG sketches, not live maps.
- The international, weighing and pallet slots use icons instead of photos, because no clean photos were supplied for them.
- The CTA band uses a photo on Home only, because each photo may be used once. The other pages use the route motif.
- The largest gap between sections' content is under the 190px limit, with little margin to spare.
- The stylesheet is about 210 KB because the fonts are inlined in it. It is cached after the first page loads.